

Foundation of Blockchain Technology (IS501) **Pre-Reading Material – Lecture 19**

Gas

Estimated Reading Time: 15–20 minutes

What is Gas?

Every interaction with the Ethereum blockchain requires **gas**.

Gas represents the computational effort required to:

- Execute operations in a smart contract.
- Process a transaction.
- Record changes permanently on the blockchain.

Simple Analogy

Think of **gas as fuel for a car**.

You need fuel to make the car move, and the amount of fuel required depends on how far or how complex the journey is.

Similarly, blockchain operations consume gas depending on the work they perform.

Why Does Gas Matter?

Gas plays an important role in Ethereum because it:

- Compensates participants for executing and validating transactions.
- Helps prevent infinite loops and wasteful computations.
- Protects the network from unnecessary or malicious use.
- Encourages developers to write **gas-efficient smart contracts**.

Three Important Gas Concepts

Term	Meaning	Simple Analogy
Gas Price	Amount of ETH paid per unit of gas	Price per litre of fuel
Gas Limit	Maximum gas units allowed for a transaction	Size of the fuel tank
Gas Cost	Gas actually consumed by an operation	Fuel consumed during a journey

Understanding the difference between these three concepts is essential for working with Solidity and Ethereum transactions.

Gas Price

Gas price is the amount a sender is willing to pay for each unit of gas.

It affects how attractive a transaction is for inclusion in a block.

Generally:

- **Higher gas price** → potentially faster inclusion.
- **Lower gas price** → potentially slower inclusion.

Analogy

If one person is willing to pay more for fuel than another, the higher-paying customer may receive priority when demand is high.

Gas Limit

The **gas limit** specifies the maximum amount of gas that the sender allows a transaction to consume.

Different operations require different amounts of gas.

If the Gas Limit is Too Low

If the transaction requires more gas than the specified limit:

- The transaction runs **out of gas**.
- The transaction fails/reverts.
- Gas already consumed during the execution attempt is not recovered.

If the Gas Limit is Sufficient

The transaction can complete successfully.

If less gas than the limit is actually used, the unused portion is returned according to the transaction's execution rules.

6. Gas Limit and Transaction Outcome

Situation	Outcome
Gas limit < required gas	Transaction fails due to insufficient gas
Gas limit is sufficient	Transaction can execute successfully
Gas used < gas limit	Unused gas is not consumed

Gas and Infinite Loops

Gas also provides protection against **infinite or excessively long computations**.

If a smart contract contains an accidental infinite loop, the execution continues consuming gas until the available gas is exhausted.

The transaction then stops rather than allowing the computation to continue indefinitely.

Key Takeaways

- **Gas = computational resource required for blockchain execution.**

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>



- **Gas price = price paid per unit of gas.**
- **Gas limit = maximum gas allowed for a transaction.**
- **Gas cost = gas actually consumed by operations.**
- Insufficient gas can cause a transaction to fail.
- Gas helps protect Ethereum from inefficient or infinite computations.
- Developers should consider **gas efficiency** when designing smart contracts.

Think Before the Lecture

1. Why does every blockchain transaction require gas?
2. What is the difference between **gas price** and **gas limit**?
3. What happens when the gas limit is insufficient?
4. Why is gas useful for preventing infinite loops?
5. Why should a Solidity developer care about gas efficiency?

